

Skills Summary

Adding, Subtracting and Multiplying Polynomials

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding polynomials

Rule: addition changes nothing inside the brackets. Drop them, then collect like terms — terms with the *same variable and the same exponent*. Only the coefficients are added: $ax^n + bx^n = (a + b)x^n$. A power that appears in only one polynomial is simply copied down.

Example: Simplify $(4x^2 + 3x - 5) + (x^2 - 7x + 2)$.

Step 1. The operation is addition, so no signs change; the only work is deciding which terms are alike.

Step 2. Squares: $4x^2 + x^2 = 5x^2$. Linear: $3x - 7x = -4x$. Constants: $-5 + 2 = -3$.
 $\Rightarrow 5x^2 - 4x - 3$

Try it: Simplify $(2x^2 + 6x) + (5x^2 - x + 4)$.

check: $7x^2 + 5x + 4$

Skill 2 — Subtracting polynomials

Rule: the minus sign belongs to *every* term of the second bracket, not just the first:

$$A - (b_2x^2 + b_1x + b_0) = A - b_2x^2 - b_1x - b_0.$$

Rewrite the subtraction as an addition of the opposite polynomial, then collect like terms exactly as in Skill 1.

Example: Simplify $(6x^2 - 2x + 1) - (2x^2 + 5x - 4)$.

Step 1. Subtraction of a whole bracket, so before collecting anything I flip every sign in the second bracket.

Step 2. $-(2x^2 + 5x - 4)$ becomes $-2x^2 - 5x + 4$; note the -4 becoming $+4$.

Step 3. Collect: $6x^2 - 2x^2 = 4x^2$, $-2x - 5x = -7x$, $1 + 4 = 5$.

$$\Rightarrow 4x^2 - 7x + 5$$

Try it: Simplify $(3x^2 + x - 6) - (x^2 - 4x + 2)$.

check: $2x^2 + 5x - 8$

Skill 3 — Multiplying polynomials

Rule: every term of the first bracket multiplies every term of the second. Multiply the coefficients and *add* the exponents: $ax^m \cdot bx^n = abx^{m+n}$. Two binomials give four products; a binomial times a trinomial gives six. Collect like terms at the end.

Example: Multiply $(x + 7)(2x - 3)$.

Step 1. Two brackets of two terms each, so I expect four products and then one pair of like terms in the middle.

Step 2. $x \cdot 2x = 2x^2$, $x \cdot (-3) = -3x$, $7 \cdot 2x = 14x$, $7 \cdot (-3) = -21$.

Step 3. Collect the middle pair: $-3x + 14x = 11x$.

$$\Rightarrow 2x^2 + 11x - 21$$

Try it: Multiply $(3x - 2)(x + 5)$.

check: $3x^2 + 13x - 10$

(!) Watch out: $(2x + 3)^2$ is *not* $4x^2 + 9$. A square is a product of two identical brackets, so it has a middle term: $(2x + 3)^2 = 4x^2 + 12x + 9$.